

Classroom

>

MAT110_SPRING_2025_SKY

20

Home

Calendar

Gemini

Enrolled

To-do

MAT110_FALL2024_SKY6

MAT110_SPRING_2025_SKY20

MAT110_SPRING_2025_SKY19

MAT110_SPRING_2025_SKY16

MAT 110(Spring-26)11

MAT110_SUMMER_2025_SKY17

MAT110_FALL_2025_SKY04

MAT110_FALL_2025_SKY03

Mat-110-section-77

MAT110 Tutorial Spring 26All

MAT 121(Spring-26)01

MAT 110Section-18

EEE101SPRING20262

EEE101 || Spring-2026Section-7

MAT216 (SPRING 2026)Sec_15 & 16

EEE 101 / ECE 101 - Spring 2026Section 4

MAT110 (Spring 2026)Section-08 and Section-16

MAT216 Spring-26 22

MAT 120Section-03

MAT 1102

MAT316 Spring-26

MAT 110(Fall-25)18

MAT 110(Fall-25)20

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

MAT110_SPRING_2025_SKY20

Upcoming

Woohoo, no work due soon!

View all

New announcement

Sakal Roy

Jan 1

HAPPY NEW YEAR $2(x^2+y^2)$.
Solve, Integers only. 😊

1 Jan 2026 at 00:55.png

Add comment

Sakal Roy

Oct 7, 2025

FYI

Faculty Initial	Days	08:00 am - 09:20 am	09:30 am - 10:50 am	11:00 am - 12:20 pm	12:30 pm - 01:50 pm
SKY	Saturday			MAT110 (3) 07H-30C	MAT110 (4) 07H-30C
	Sunday				
	Monday				MAT456 (1) 10H-39C
	Tuesday				
	Wednesday				MAT456 (1) 10H-39C
Thursday				MAT110 (3) 07H-30C	MAT110 (4) 07H-30C

SKY_FALL_2025.png

Add comment

Sakal Roy

Aug 12, 2025 (Edited Aug 12, 2025)

@everyone
Dear Students,
Join me in the talk 14 august, 5:00 pm, if you have time. See you all.
<https://www.facebook.com/share/p/1AaTUK8k65/>

Add comment

Sakal Roy

Jul 18, 2025

@everyone
INVITATION TO A JOURNEY
ITS THE TIME OF THE YEAR AGAIN...
Hello Math lovers,
Probably you know that in FALL 2024 I taught "MAT 456: TOPICS IN COMBINATORICS" at MNS, BRACU .This was a brand new elective course (IF math major/minor THEN OK, ELSE please check with respective departments) and was the first solely dedicated course on COMBINATORICS in BRACU. Now, I wish to teach the course again in next semester FALL 2025 in MNS, BRACU.
ARE YOU READY? THEN SHOW YOUR WISH IN WISHLIST.

Here's just a tentative brief design of the course and detailed assessments from FALL 2024:
<https://shorturl.at/beim5>

That's just a brief, more to be unfolded. And, I believe in collaboration, so keeping this document open for any comment.
LET'S COMBINE OUR THOUGHTS...
<A fun poster is attached for VIBECHECK of the course. Surely I don't look like this, so poster credit/blame goes HALF to AI and HALF to HAFIZ, my beloved brother in crime and colleague.>

Add comment

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

1/17

Best wishes.

•
•
•

▷

•
•
•

...

▷

...

27 MAY, 2025 (TUE) FROM 2:00 PM- 5:00 PM



Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

MAT110_SPRING_2025_SKY 20

QUIZ, ASSIGNMENT, ATTENDANCE: MARKS WILL BE GIVEN IN FULL GRADESHEET along with MID and FINAL marks.

**BRACU HAS DECIDED THAT :THIS SEMESTER ASSIGNMENTS ARE ALSO BEING EVALUATED BY COURSE FACULTY*

Stream

Classwork

People

1 class comment

Sakal Roy • May 25, 2025

Emailing me is sufficient.

P:S: Your friend has to nominate you to give consent. Not the other way around.

Add class comment...

Sakal Roy

May 23, 2025

Dear All,
Your MID TERM SCRIPT has been assessed using the attached marking scheme and detailed solution.

- I'm trying to do the same for FINAL Scripts. But proobably won't be able to provide detailed solution like MID TERM one you're seeing here due to time constraint.
- Will announce for MID and FINAL SCRIPT CHECKING whenever everything is finished.
- Here is the gist of the ASSESSMENT STRATEGY I follow. You'll get this break down also while receiving MID and FINAL marks in individual email.

SOLUTION & MARKING SCHEME(Created by: SAKAL ROY [SKY])

TO APPRECIATE YOUR MATHEMATICAL THINKING, ANY VALID MATHEMATICAL WAY OF SOLVING WILL BE TREATED EQUALLY AND ASSESSED ACCORDINGLY, AS LONG AS IT'S CONSISTENT WITH THE PROVIDED INSTRUCTIONS OF THE QUESTION.

IN THIS ASSESSMENT PROCESS EVERY PROBLEM WILL BE ASSESSED BASED ON 3 CRITERIA:

A. MATHEMATICAL THINKING

B. MATHEMATICAL KNOWLEDGE

C. MATHEMATICAL SKILL

Hope that will help you understanding your performance even better and in a more sophisticated manner. Best wishes

MID_TERM_MAT_110_SPRIN...

PDF



Add comment

Sakal Roy

May 22, 2025

Hello everyone.
Hope you are all well.

- I'll soon make an announcement regarding marks of MID, script checking date and assignment ,QUIZ marks first.
- Then afterwards FINAL marks and scripts checking the date of final scripts will be given.
- FINAL marks will take time to ensure that you all get what you deserve with appropriate marking scheme, solution and partial marking criteria.

Thanks in advance for your patience.
Best wishes.

Add comment

Sakal Roy

May 19, 2025

Recording link:
<https://bdren.zoom.us/rec/share/OG5sHA-KRmwetce1ZM5snWZUVH-Vn9qhaLDgg9tzl-SWJtu6yBAIg1RDEq22vYs.z2H8M9mas9YhMIYe>

Drawing board (Use Draw.io to view it):
<https://drive.google.com/file/d/15dt2BMnRfHbVOp5XiKQeajug9brucBdF/view?usp=sharing>

Best wishes.

Error - Zoom

<https://bdren.zoom.us/rec/share/OC>



MAT 110- FINAL: TIPS AND ...

Unknown



Add comment

Sakal Roy

May 18, 2025

Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: MAT 110- FINAL: TIPS AND TRICKS
Time: May 18, 2025 08:00 PM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lboJ0I9.1&omn=95711536818>

Meeting ID: 231 308 1916
Password: savelife

=====

- See previously shared MSG from Course Coordinator here again:**

=====

Syllabus for Final Exam:

Add comment

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

4/17

Home

Calendar

Gemini

Enrolled

To-do

MAT110_FALL2024_SKY6

MAT110_SPRING_2025_SKY20

MAT110_SPRING_2025_SKY19

MAT110_SPRING_2025_SKY16

MAT 110(Spring-26)11

MAT110_SUMMER_2025_SKY17

MAT110_FALL_2025_SKY04

MAT110_FALL_2025_SKY03

Mat-110-section-77

MAT110 Tutorial Spring 26All

MAT 121(Spring-26)01

MAT 110Section-18

EEE101SPRING20262

EEE101 || Spring-2026Section-7

MAT216 (SPRING 2026)Sec_15 & 16

EEE 101 / ECE 101 - Spring 2026Section 4

MAT110 (Spring 2026)Section-08 and Section-16

MAT216 Spring-26 22

MAT 120Section-03

MAT 1102

MAT316 Spring-26

MAT 110(Fall-25)18

MAT 110(Fall-25)20

MAT 122(Fall-25)01

Archived classes

Settings

StreamClassworkPeople

- MAT110_SPRING_2025_SKY 20
- Partial derivatives
 - Chain rules for partial derivatives
 - Maxima and Minima of functions of two variables



Some sections (Parabola, ellipse and hyperbola),

- Conversion of points (Rectangular, Cylindrical and Spherical Coordinates)

Note: Students will answer 5 questions out of 6 questions.Marks: 7*5=35
Exam Duration: 1 hr 20 min
=====

Discalimer: Just solve the practice problems and related examples or final. Because, as you've already seen ***in the previously shared MSG from Course Coordinator:These topics have been introduced in class or discussed but mentioned as **not to appear in final exam:*****

- Partial derivatives:
Implicit Differtiation,
- Chain rules for partial derivatives
- Maxima and Minima of functions of two variables:
Absolute extrema (maxima/minima) and Optimization problems
- Lagrange Multipliers to find maximum and minimum values :
Optimization problems
- Taylor expansion (Linear and Quadratic) of functions of two variables:
3rd or higher degree approximation, 3 variable function.
- Conic sections (Parabola, ellipse and hyperbola)
- Conics in polar coordinates
- Conversion of points (Rectangular, Cylindrical and Spherical Coordinates)

2 class comments

Sakal Roy • May 18, 2025

All things will appear in final from conversion of points (rectangula, cylindrical and spherical points)
That means no things are exclude (that's why its kept blank).
[Just like in "chain rule for partial derivatives" you need all of it. So nothing is mentioned as excluded there.]

Add class comment...

Sakal Roy

May 16, 2025

Dear All,

You do not need to show proof of the formulas. Just using them 100% correctly is sufficient for full marks. However, knowing tiny details about the formulas and a brief idea about working steps greatly helps avoid mistakes. **Intuition combined with observation works as a shield against common errors.**

https://bdren.zoom.us/rec/share/SdtzaGntnTgEJFrusMFFvXYKnzjZOb2nRkdYPwxaDWORoRPJPVFbhbM3Az9lowSj_9PR8qvvQmPzLB-j

BEST WISHES.

Error - Zoom

https://bdren.zoom.us/rec/share/Sc

Add comment

Sakal Roy

May 15, 2025

Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: MAT 110-CONICS

- Conic sections (Parabola, ellipse and hyperbola)
- Conversion of points (Rectangular, Cylindrical and Spherical Coordinates)

Time: May 16, 2025 10:00 AM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lboJJOi9.1&omn=94659374493>

Meeting ID: 231 308 1916
Password: savelife

(Recording will be provided immediately after the session, feel free to drop your query beforehand here in comments and in email also if you prefer that. Remember not everything from conics mentioned in course outline are in FINAL EXAM. See Course ordinators msg shared here earlier to know exactly which parts are included. I'll make that clear in this session also if you've confusion.)

Add comment

Sakal Roy

May 13, 2025

Recording:

Topics:

1) Maxima minima of two variable functions

2) Lagrange multiplier for finding maximum and minimum values of multivariable functions

https://bdren.zoom.us/rec/share/f2sRyzg3YM8yx3vCEamVmRjD2Bp9LPNSS6dAL4105YByfOyRj5yJ1dMzp5A6cF_B.NZyKfW5i9TnFSAA6

Error - Zoom

https://bdren.zoom.us/rec/share/f2:

Add comment

Sakal Roy

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

?

5/17

Home

Calendar

Gemini

Enrolled

To-do

MAT110_FALL2024_SKY6

MAT110_SPRING_2025_SKY20

MAT110_SPRING_2025_SKY19

MAT110_SPRING_2025_SKY16

MAT 110(Spring-26)11

MAT110_SUMMER_2025_SKY17

MAT110_FALL_2025_SKY04

MAT110_FALL_2025_SKY03

Mat-110-section-77

MAT110 Tutorial Spring 26All

MAT 121(Spring-26)01

MAT 110Section-18

EEE101SPRING20262

EEE101 || Spring-2026Section-7

MAT216 (SPRING 2026)Sec_15 & 16

EEE 101 / ECE 101 - Spring 2026Section 4

MAT110 (Spring 2026)Section-08 and Section-16

MAT216 Spring-26 22

MAT 120Section-03

MAT 1102

MAT316 Spring-26

MAT 110(Fall-25)18

MAT 110(Fall-25)20

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

MAT110_SPRING_2025_SKY 20

May 13, 2025

• **ANNOUNCEMENT** : ANNOUNCING ONLINE CLASS ON 13 MAY, 2025 (TUESDAY), 00:00 PM

Topic: MAT 110

1) Maxima and Minima of multivariable functions

2) Vector Calculus (Gradient) and Lagrange Multipliers

Time: **May 13, 2025 08:00 PM** Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lbolJOI9.1&omn=98044218261>

Meeting ID: 231 308 1916
Password: savelife

Add comment

Sakal Roy

May 12, 2025

Dear All,

Here are minimum resources for **lagrange multiplier for self study first**.

Also, I'm sharing only the excercises you need to practise and related online solvers here.

Make sure you see only see and practise them and nothing more since EXAM is knocking at the door.

Keep in mind that, we've got less offline classes in this semester that ever due unfortunate events. And that's why everything is not in this semesters scope. So, utilize your time properly.

I'll announce 1/2 online sessions afterwards including an exam review class before final.

So, email me your concerns beforehand in this format so that I can discuss in review sessions accordingly:

=====

STUDENT ID:

FULL NAME:

TOPIC NAME:

Reference RESOURCE that you've gone through :

(mentioned/mark specifically where you're facing problems, mention time interval if its a video)

EXAMPLE/EXCERCISE NO:

PROBLEM NO:

=====

Practise: **HOWARD ANTON : 13.9: Lagrange Multipliers: 5-12**

5–12 Use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint. Also, find the points at which these extreme values occur. ■

5. $f(x, y) = xy$; $4x^2 + 8y^2 = 16$

6. $f(x, y) = x^2 - y^2$; $x^2 + y^2 = 25$

7. $f(x, y) = 4x^3 + y^2$; $2x^2 + y^2 = 1$

8. $f(x, y) = x - 3y - 1$; $x^2 + 3y^2 = 16$

9. $f(x, y, z) = 2x + y - 2z$; $x^2 + y^2 + z^2 = 4$

10. $f(x, y, z) = 3x + 6y + 2z$; $2x^2 + 4y^2 + z^2 = 70$

11. $f(x, y, z) = xyz$; $x^2 + y^2 + z^2 = 1$

12. $f(x, y, z) = x^4 + y^4 + z^4$; $x^2 + y^2 + z^2 = 1$

Lagrange.png

Lagrange Multipliers | Geo...

YouTube video • 12 minutes

Lagrange Multipliers Calcul...

<https://www.emathhelp.net/calculators>

Add comment

Sakal Roy

May 12, 2025 (Edited May 12, 2025)

Dear All,

- I'm cancelling todays consultation session.
- And announcing a **MEGA Consultation hour** on:
15 may, 2025 (THU) from **12:00 PM-6:00 PM**
- Also, I wish to take 2 online session +1 exam review session (recording will be provides).**

Send your query/confusion in email beforehand.

Take care.

All the best.

Add comment

Sakal Roy

May 12, 2025

Hello everyone,

See the problem in the attachment.

Try to solve it yourself in the first 15 minutes, write your comments about it.

But but no discussion is allowed with others.

After 15 minutes, I'll then start the class by checking to your thought and ideas.

Best wishes.

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

6/17

Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

Problem:

$$a \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + b \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 10 \\ 3 \\ 8 \end{pmatrix}$$

Solve for the unknowns a and b .

12 May 2025 at 10:55.jpg

Add comment

Sakal Roy

May 8, 2025

- Maxima and Minima of multivariable functions:**
#See the attached video to get an idea about exactly what we need to do.
#See from google doc the examples and solve practice problems:
HOWARD ANTON : 13.8: Maxima and Minima of Functions of two Variables: 9-19#Use Solver to cross check your soution.
- Discalimer:** Just solve the practice problems and related examples or final. Because, as you've already seen ***the previously shared MSG from Course Coordinator:****Absolute extrema (maxima/minima) and Optimization problems are not recommended in this semesters final outline.*
=====
- See previously shared MSG from Course Coordinator here again:**
=====

Syllabus for Final Exam:

- Partial derivatives
- Chain rules for partial derivatives
- Maxima and Minima of functions of two variables
- Lagrange Multipliers to find maximum and minimum values
- Taylor expansion (Linear and Quadratic) of functions of two variables
- Conic sections (Parabola, ellipse and hyperbola)
- Conversion of points (Rectangular, Cylindrical and Spherical Coordinates)

Note:Students will answer 5 questions out of 6 questions.Marks: 7*5=35Exam Duration: 1 hr 20 min=====

Multi-variable Optimization...

YouTube video • 13 minutes



Critical Points, Extrema, an...

https://www.emathhelp.net/calcula:



Add comment

Sakal Roy

May 6, 2025

For Exam preparation regarding Taylor Polynomial of functions of two variables read only this resource (practise problems and resource links can be found in google doc also).
See you all in next class.

Taylor Polynomials of Funct...

https://math.libretexts.org/Bookshe



Add comment

Sakal Roy

Apr 30, 2025

Will start class 10 minutes later discussing problemsolving techniques and tips regarding chain rule for multivariable calculus.
Then, we'll explore Taylor approximation.
Resources are given in Course info google doc for spring 2025.

Course Info: MAT_110_SKY_...

Google Docs



Add comment

Sakal Roy

Apr 26, 2025

Dear All,
In case you haven't submit your assignment within the deadline, do submit it within **monday**.
Though that's in the late submission period, but I'll request your ST to consider that if you email me mentioning your reason for missing the extended deadline.
Take care. Best wishes.
See you all in next class, don't miss it.

Add comment

Sakal Roy

Apr 23, 2025

Dear students,
To finish PARTIAL DERIVATIVE and CHAIN RULE for PARTIAL DERIVATIVES once and for all:

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

7/17

Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

MAT110_SPRING_2025_SKY 20

- Start with watching the resources attached here.
- Then Read directly from **GOOGLE DOC** to understand selected parts from **HOWARD ANTON**.
- After you're confident practice **BRACU** suggested **practice sheet problems and excercises** from **GOOGE**

See you all in next class.

Best wishes.

FUN FACT:

WHO'LL EAT WHAT?

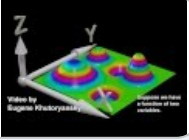


Believe it or not, Solving this puzzle helps drawing tree diagrams for Chain Rule for Partial Derivatives.

FOOD PUZZLE.png

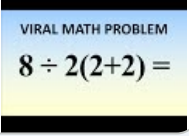
Gradients and Partial Deriv...

YouTube video • 5 minutes



8÷2(2+2) = ? Mathematicia...

YouTube video • 5 minutes



The Multi-Variable Chain R...

YouTube video • 10 minutes



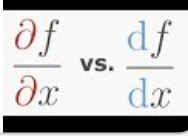
What are derivatives in 3D?...

YouTube video • 8 minutes



Difference Between Partial ...

YouTube video • 1 minute



Course Info: MAT_110_SKY_...

Google Docs



Add comment



Sakal Roy

Apr 22, 2025

Dear All,

Greeting, hope you're enjoying boisakh celebration at BRACU.

I'll continue consultation after lunch break on 2:30 pm instead of 2:00 pm. Will continue till 5:30 pminstead of 5:00 pm.

Email me personally with your queries beforehand to ease the consultation process. Also, don't worry there'll be more mega consultation hours like this in next weeks. Best wishes.

Add comment



Sakal Roy

Apr 20, 2025

Dear All,

Since, today 20 april is Easter Sunday and as I came to know BRACU is going to celebrate "Pohela Boisakh" on 22nd april. So, for everyones convenience, I'm extending the assignment deadline upto 25 April, 2025 (FRIDAY) 11:59 PM.

Don't miss the extended deadline. Submission's will be counted as LATE after the extended deadline.

Best wishes.

Enjoy.

Add comment



Sakal Roy

Apr 19, 2025

Dear All,

Announcements:

- No offline class or consultation will be from me on

21 APRIL, 2025 (MON).

- I'm announcing a 6 hour long - MEGA CONSULTATION HOUR ON:
22 APRIL, 2025 (TUES) from: 10:00AM - 1:00 PM and 2:00 PM-5:00 PM.

Instructions:

- First see my **Recorded Class from MAT 110-sec 16 (1 hour 30 minutes)**.
- Then see the attached **short videos (8 mins in total)** to explore further, those will likely to be appear in your next assignments/presentation/groupwork/project for MAT 110. I'll announce them after you complete submission of your first assignment in due time. Hopefully that'll help you focusing one thing at a time.

BE SURE TO CLEAR ALL YOUR CONFUSION AND DON'T IGNORE ANY MATH EMERGENCY.

MAY THE BEAUTY AND JOY OF MATHEMATICS GUIDE YOU TO SUCCESS.

BEST WISHES.

Meeting Summary for MAT 110:

AI-generated content may be inaccurate or misleading. Always check for accuracy.

Apr 19, 2025 12:29 PM Astana, Dhaka ID: 231 308 1916

Quick recap

Sakal introduced the concept of partial derivatives and change rules, focusing on problem-solving and visualization, and explained the basics of functions with single and multiple variables. He also discussed the domain, codomain, and range of functions, and the behavior of these functions for specific inputs, including zero and negative numbers. Additionally, Sakal explained the concept of Cartesian product in mathematics, using a deck of cards as an example, and emphasized the importance of understanding domains and co-domains in functions.

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

8/17

Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

Next steps

- Students to review the Google Classroom materials on coordinate systems and multi-dimensional visualization
- Students to complete the assignment on graphing multi-variable functions.
- Sakal to provide more detailed examples and practice problems in the next session.

Summary

Partial Derivatives and Function Basics

Sakal introduces the concept of partial derivatives and change rules, focusing on problem-solving and visualization. He explains the basics of functions with single and multiple variables, using examples to illustrate domains and codomains. The discussion covers the notation for functions, including input and output variables, and emphasizes the importance of clear understanding in mathematical modeling. Sakal also mentions that students will need to plot functions and create short videos as part of their assignments or group projects.

Functions in Mathematics: Domain, Codomain, Range

Sakal explains the concept of functions in mathematics, using countries and their capitals as examples. He clarifies that while the relationship between countries and capitals is a relation, it's not necessarily a function due to the one-to-one correspondence requirement. Sakal then discusses the domain, codomain, and range of functions, emphasizing that the range consists of actual outputs, which may be a subset of the codomain. He uses examples like Bangladesh, Nepal, and China to illustrate these concepts.

Domain and Range of Mathematical Functions

Sakal discusses the domain and range of various mathematical functions, focusing on real and complex numbers. He explains that for the function $f(x) = x^3$, the domain and range are all real numbers. For $f(x) = x^2$, the domain is all real numbers, but the range is limited to non-negative real numbers. Sakal also explores the behavior of these functions for specific inputs, including zero and negative numbers, and discusses the concept of infinity in relation to real numbers.

Multivariable Calculus: Understanding Domains and Co-Domains

Sakal discusses the methodology for dealing with negative Bose and introduces the concept of multivariable calculus. He explains the importance of understanding domains and co-domains in functions, emphasizing the need to clearly define input variables. Sakal demonstrates how to plot functions using visualization tools and calculators, encouraging students to practice writing functions with multiple input variables. He stresses the significance of recognizing independent variables and understanding the set of valid inputs in multivariable calculus.

Cartesian Product in Mathematics Example

Sakal explains the concept of Cartesian product in mathematics, using a deck of cards as an example. He describes how ordered pairs (x,y) represent elements in the Cartesian product of two sets, where x belongs to set A and y belongs to set B . Sakal emphasizes the importance of order in these pairs and demonstrates how this concept applies to a standard deck of playing cards, where suits (hearts, spades, clubs, diamonds) form one set and card values (ace, 2, 3, ..., king) form another.

Error - Zoom

https://bdren.zoom.us/rec/share/Ff.

CalcBLUE 2 : Ch. 1.2 : Visual...

YouTube video • 2 minutes

CalcBLUE 2 : Ch. 1.4 : Multiv...

YouTube video • 4 minutes

CalcBLUE 2 : Ch. 1.5 : Towar...

YouTube video • 0 minutes

2 class comments

Sakal Roy • Apr 20, 2025

Dear Shatvik:
Nope, I don't have any class/consultation today, so probably won't be at BRACU . You can email be your query. Or join me in the mega consultation hour on tuesday. Best wishes.

Add class comment...

Sakal Roy

Apr 19, 2025

Dear All,
You've already received email from BRACU regarding online class on 19th APRIL, 2025 (SAT).

- So, there'll be No consultation for 19 April, 2025 (SAT) since it's not possible to take offline.***
- However, I'll announce offline MEGA CONSULTATION HOUR later to compensate that.***

Email me your queries in the meantime .

Add comment

Sakal Roy

Apr 16, 2025

Hello, everyone. Hope you are doing well.
I will start todays class officially at 20 minutes later.

Todays task is to:

#THINK ALL BY YOURSELF
(first 20 minutes on your own)

#THINK HARDER TOGETHER
(when class officially begins)

These first 20 minutes are for your self exploration regarding the following questins:

#How do human see?
#Do you think we can see 3D?

Write down your vote YES/NO and corresponding ideas, reasoning.

I'll listen to them while starting the class.

6 class comments

NURUNNOBI CHOWDHURY • Apr 16, 2025

No.

Add class comment...

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

9/17

Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Sakal Roy

Apr 12, 2025

Stream

Classwork

People



online.

So, I won't take any class today. But, I am shaing the resources for next class beforehand. You'll also find them in course info Googledoc for SPRING 2025.

Feel free to go through them, explore and experiment.

In next class we'll enjoy the tale of real life and mathemaics together and get ready to apply those experiences later.

Also, no consultation for today, send me your query in email if necessary. I'll respond there accordingly.

I wish to announce A day long consultation on next week to help you all preparing.

Best wishes.

Hole in Your Hand Illusion : ...

https://www.instructables.com/Hole-in-Your-Hand-Illusion

CalcBLUE 2 : Ch. 1.1 : Graph...

YouTube video • 1 minute

Cartesian product - Wikipe...

https://en.wikipedia.org/wiki/Cartesian_product

CalcBLUE 1 : Ch. 3.1 : Coord...

YouTube video • 2 minutes

Add comment

Sakal Roy

Apr 11, 2025 (Edited Apr 11, 2025)

MSG FROM COURSE-COORDINATOR:
=====

Dear esteemed colleagues,
We are going to cover the following syllabus for the final exam.

• Partial derivatives

• Chain rules for partial derivatives

• Maxima and Minima of functions of two variables

• Lagrange Multipliers to find maximum and minimum values

• Taylor expansion (Linear and Quadratic) of functions of two variables

• Conic sections (Parabola, ellipse and hyperbola)

• Conversion of points (Rectangular, Cylindrical and Spherical Coordinates)

=====

Add comment

Sakal Roy

Apr 9, 2025 (Edited Apr 9, 2025)

Hello, everyone. Hope you are doing well.
Will start todays class at 25 minutes later.
Todays class consists of discussing the basic ideas needed to move forward.
Also we'll discuss what is multivariable calculus and selected topics from multivariable calculus for for MAT110 in this semester SPRING 2025.
Will listen to how your EID vacation went and get your opinion on the given movie/ movie clip.
Will also discuss about any query regarding the announced written assignment.

Independent and Dependence...

YouTube video • 6 minutes

Explicit Function - Meaning...

https://www.cuemath.com/algebra/explicit-function

Implicit Function - Definitio...

https://www.cuemath.com/algebra/implicit-function

What are the big ideas of M...

YouTube video • 16 minutes

Add comment

Sakal Roy

Apr 6, 2025

By this time, you all are supposed to receive an official announcement from BRACU in your individual bracu mail. Sharing here again for convenience. Before that read here a clarification from me besides official msg from BRACU.

My MSG for you:

Thanks for reaching out to me requesting tomorrows stance of solidarity with gaza.

As you've already noticed in the earlier announcement in the google classrooms of my sections (MAT 11

• Osec 16,19,20), there is no offline class from me on 7th April,2025 (MONDAY). But to make it clear, I announced it earlier so honestly speaking it is not linked with attack on Gaza.

However, even though I announced for no class on 7th, I wanted to announce extra consultation hour. But appreciating your stances for humanity, I won't announce that on 7th anymore.

Let's just hope that your collective stance will cease the war. Though its not anymore a war, its plain and simple genocide.

Regards

Sakal Roy

Adjunct (Contractual) Faculty,

MNS, BRACU

Official msg from BRACU:

Dear Students, Faculty, and Staff,

We are deeply concerned about the ongoing crisis in Palestine and stand in solidarity with those affected by these tragic events. Our thoughts and prayers are with all who are enduring suffering, and we support the universal call for peace, justice, and human dignity.

In support of this cause, we uphold our students' right to express their solidarity in peaceful and meaningful ways. Therefore, no academic activities will take place tomorrow; students at RS will receive specific guidance from RS management.

We are with you during this challenging time. Following tomorrow, all academic activities will resume as scheduled, since we are mindful of the need to avoid losing any valuable teaching time.

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

10/17

Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Thank you for your understanding and commitment to our community values during these difficult times.



Office of the Registrar
BRAC University

1 class comment

YASIR ARAFAT • Apr 6, 2025
Thank you for the support, sir.

Add class comment...

Sakal Roy
Apr 4, 2025 (Edited Apr 4, 2025)

Hello, hope you're all enjoying your vacation.
I promised 3 things to you earlier:

- ✔ Movie Day! – we'll watch movie in the first classes after Eid.
 - ✔ No new topics in the first week – We'll focus on self-reflection and self-assessment of midterm topics.
 - ✔ Assignment flexibility – You can use books, online resources, even ChatGPT (with proper reference), but no human help to avoid violating BRACU plagiarism policy.
- So, lets see what I can do to keep my promises.

🔥 **Movie Assignment – Watch 2 movies (clips attached for reference).**

Then Think about:
→ What's the connection between this course and these movies?
→ Share your thoughts in our next in-person class at BRACU.

🔥 **Midterm Topics Assignment** – See attached.

Announcement on next week's Class Schedule:

📺 **Movie Day (No Regular Class! No Online/Offline Class! Watch movie at home,that's the first part of the movie assignment):**

Sec-16: April 5, 2025 (Sat)

Sec-19, 20: April 7, 2025 (Mon)

💡 **Discussion on the Movie and midterm Assignment (AT BRACU in regular class):**

Sec-16: April 10, 2025 (Thu)

Sec-19, 20: April 9, 2025 (Wed)

⚠️ **Assignment is announced early – Read carefully & avoid late submission penalties!**

🔗 Google Doc Version (navigate using index):

https://docs.google.com/document/d/1K7guG5wcWxnY1AiktAjRfiUIGZ_VZGpHlqiEydjQLaM/edit?usp=drivesdk

Or

📄 PDF version: (download and then navigate):

https://drive.google.com/file/d/13Nsl_KPclrU3sEz9eZZwPsAMZMfJHz6Y/view?usp=drivesdk

Final Notes

✈️ Travel safely back to Dhaka! Looking forward to hearing your Eid stories and diving into more fun with Calculus. Best wishes!

MAT110_SKY_SPRING-2025...

Google Docs



MAT110_SKY_SPRING-2025...

PDF



Mission Impossible: Ghost ...

YouTube video • 7 minutes



Pettigrew on The Marauder'...

YouTube video • 4 minutes



 Add comment

Sakal Roy
Mar 27, 2025

Hello, everyone.
Hope you're doing fine.
There'll be no class from me before EID-UL-FITR.
So, now, I've 3 important news to share:

1) I'll announce an online assignment on MID TERM TOPICS. You'll have 2 weeks(14 days) to submit it. But no need to worry about that now.
I'll announce it after EID VACATION ends.

2) We'll continue our unfinished Team Quiz in the first week after EID. And I'm thinking of planning some sort of celebration (not only for the winning team but also the whole class). So, get ready for a convenient mathematical planing for that too.

3) Enjoy the EID FESTIVAL with your family and friends. Be ready to share your experiences.

EID MUBARAK IN ADVANCE.

Stay safe, take care.
Best wishes.

 Add comment

Sakal Roy
Mar 18, 2025



Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

MAT110_SPRING_2025_SKY 20

Prepare well, stay cool while answering the exam questions, read properly, write everything that's going on your head.

I won't mind even if that's any song or painting. After all Mathematics is an art, and you are the artist here.

Stream

Classwork

People

91NKix-fc1K70Sbn.SagxpmRjiE-rtUXl?accessLevel=meeting&canPlayFromShare=true&from=my_recording&continueMode=true&componentName=rec-play&originRequestUrl=https%3A%2F%2Fbdren.zoom.us%2Frec%2Fshare%2FNI-F4leNcKPST7MY2WdXJTTTklcaxvDEYcQkz5nwjcreYCPYFOqyeiOdbj1Z6cBy.7lQwg5L-WYk2qdYp

Error - Zoom

https://bdren.zoom.us/rec/play/CQ!

2 class comments

Sakal Roy • Mar 19, 2025

Mat 110 is a Central exam. And central exam times don't usually change without course coordinator, departments permission, that's also in case of severe emergency situation. Till now, I haven't received any msg regarding such till now. So, Prepare according the routine, check your sections and exam room, time carefully. N:B: Usually non-central course might change time easily. You may check if CSE 111 is a non-central course. Let me know if you face any kind of overlapping situation for CSE 111 schedule changes.

Add class comment...

Sakal Roy

Mar 18, 2025

Announcement:

I'll be in MNS BRACU 4th floor , usual consultation place from **3:30 PM- 5:00 PM.** Meet me in case of any MATH EMERGENCY. Since, usually there is no consultation available in teachers zone during exam time. So, Email me your query before coming. Hope that helps. Sakal Roy

Add comment

Sakal Roy

Mar 18, 2025

MAT110_MIDTERM_FAQ_SPRING 2025_SKY

https://docs.google.com/document/d/1IHVQQyUI9BOzXx1ToSMgq2mIJ3C_bkN926_VTK7VAOQ/edit?usp=sharing

📌📌 Collected from the comments in Google Classroom and emails (Sections 16, 19, 20) Thanks to the students who asked these questions—you should thank them too!

Keep asking more questions and I'll add those here. These are quick responses so if you find any discrepancy here, let me know. You can Comment/Suggest in the FAQ google doc directly, or feel free to email me. I'll try to keep updating ASAP. For any confusion or detailed answer please refer to the main course info google doc.

📌📌 Reminder: Check the Google Classroom for updates, and refer to the Google Doc & recorded review sessions for detailed explanations. 🌟 Good luck with your midterms!Best wishes,Sakal Roy

Add comment

Sakal Roy

Mar 17, 2025

Recording of last 2 review sessions:https://bdren.zoom.us/rec/share/FQ1jnga7_URacvEgn0sgPJ68woL_D_lIFLkCghwThfh9uby5aSoHGBKJ8bJW2tV8.HUQzuyCii9ZDr7cYhttps://bdren.zoom.us/rec/share/gkCwnFpGh0dkkXFTuOnHJnFXy84UH5VG0g58Yovnlx7G3Qj7LuOPP1J7CUCrYCDEj2HdbkjZxkE3EUWa

Drawing board for last 2 review sessions:https://drive.google.com/file/d/1n3dh_Ai8d0YVHrrQlOCaCaFBSz0aGS7h/view?usp=sharinghttps://drive.google.com/file/d/1wAujHVaB_olz_QpaMO-iXbkAKCUuWDYv/view?usp=sharing

Try the BRACU suggested practice problems first.

Read only from google doc and use given solvers to prepare smartly.

I wish to take an ONLINE EXAM STRATEGY CLASS on 18 march, 2025 (TUE) at 7:30 PM

Joining link:
Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: ALL EYES ON_MAT110_MID TERM
Time: Mar 18, 2025 07:30 PM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lbolJOI9.1&omn=97650930677>

Meeting ID: 231 308 1916
Password: savelife

BEST WISHES.

Error - Zoom

https://bdren.zoom.us/rec/share/FC

MAT 110_SKY_EXAM REVIE...

Unknown

Error - Zoom

https://bdren.zoom.us/rec/share/gk

MAT 110_SKY_MID TERM EX...

Unknown

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

12/17

Home

Calendar

Gemini

Enrolled

To-do

MAT110_FALL2024_SKY6

MAT110_SPRING_2025_SKY20

MAT110_SPRING_2025_SKY19

MAT110_SPRING_2025_SKY16

MAT 110(Spring-26)11

MAT110_SUMMER_2025_SKY17

MAT110_FALL_2025_SKY04

MAT110_FALL_2025_SKY03

Mat-110-section-77

MAT110 Tutorial Spring 26All

MAT 121(Spring-26)01

MAT 110Section-18

EEE101SPRING20262

EEE101 || Spring-2026Section-7

MAT216 (SPRING 2026)Sec_15 & 16

EEE 101 / ECE 101 – Spring 2026Section 4

MAT110 (Spring 2026)Section-08 and Section-16

MAT216 Spring-26 22

MAT 120Section-03

MAT 1102

MAT316 Spring-26

MAT 110(Fall-25)18

MAT 110(Fall-25)20

MAT 122(Fall-25)01

Archived classes

Settings

- Stream
- Classwork
- People



2 class comments

See from the resources shown in record/given in google classroom or google doc directly.

<https://www.mathsisfun.com/calculus/l-hopitals-rule.html>

In short:
la hopita's law can be used in determining $\lim_{x \rightarrow a} [f(x)/g(x)]$ only if $f(x)/g(x)$ goes to Indeterminate form as $x \rightarrow a$.
And also keep in mind the cases where Lahopitas law fails:
1) if $f(x)$ or $g(x)$ is not differemtible at $x=a$
2) if $\lim_{x \rightarrow a} [f(x)/g(x)]$ does not exists(DNE)

3) if applying law hopitas law leads to circular reasoning
In google doc this type of example where applying lahopitas is not allowed is given as well as it is mentioned in the record of review session 1.



Add class comment...



Sakal Roy
Mar 16, 2025



If you don't find me at my desk for bonus consultation today (9:00 AM-12:00 PM), Please, email me. I'm at 4th floor, MNS but in another official meeting right now.
Let me know if any of you come and I'll address you. Thanks for your understanding and patience.

Add comment



Sakal Roy
Mar 15, 2025



4 URGENT ANNOUNCEMENTS:
1) Regarding Topics for MID TERM (confirmed from Course Coordinator):

- Limit and Continuity,
- Differentiability,
- Techiques of Differentiation,
- Rolle's and Mean value theorem,
- Maxima-minima
- Taylor and Maclaurin series/polynomial

That means:
1) **Succesive Differentiation and LEIBNITZ theorem** will not be included in MID TERM EXAM this semester.
2) In maxima-minima you jusst need to know how to find extrema(maxima-minima), increasing-decreasing interval,concave up-down interval. You don't need to take preparaion for **OPTIMIZATION PROBLEMS or REAL LIFE SCENARIO** parts from there for MID-TERM EXAM.
But, remember they'll be needed in FINAL EXAM TOPICS and also in your next courses. So, I'll obviously discuss them as recap in the classes after MID TERM EXAM. Just don't worry about them before MID TERM EXAM.

2) Regarding Question Pattern for MID TERM EXAM:
Tentative number of question: 6 ,
You've to answer 5 question. Each question will have 8 marks.
So, in total $5 \times 8 = 40$ marks. The obained mark out of 40 will be converted into out of 20.

3) Regarding MAT 110_SKY_MID TERM EXAM REVIEW SESSION-2:Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: MAT 110_SKY_MID TERM EXAM REVIEW SESSION-2
Time: Mar 16, 2025 08:00 PM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdyxs9lbolJOI9.1&omn=96370239063>

Meeting ID: 231 308 1916
Password: savelife

Recording will be given as soon as its available.

4) Regarding BONUS COSULTATION HOUR:
Since, 17 march,2025 (MON) all academic activities are off as per BRACU CALEDAR.
So the scheduled consultation will not be possible on that day.
Therefore, to aid your exam preparation, I'm announcing a

BONUS COSULTATION HOUR on:
Tomorrow 16 March, 2025(SUN) from 9:00 AM- 12:00 PM
However, feel free to email your quey/confusion here in GOOGLE CLASSROOM or in my official email in case you can't join the consultation hour.
Best Wishes.

Add comment



Sakal Roy
Mar 13, 2025



Recording of todays review session:

https://bdren.zoom.us/rec/share/FQ1jnga7_URacvEgnOsgPJ68woL_D_IIFLkCghwThfh9uby5aSoHGBKJ8bJW2tV8.HUOzuyCii9ZDr7cY

Error - Zoom

<https://bdren.zoom.us/rec/share/FC>



Add comment



Home

Calendar

Gemini

Enrolled

To-do

M

MAT110_FALL2024_SKY6

M

MAT110_SPRING_2025_SKY20

M

MAT110_SPRING_2025_SKY19

M

MAT110_SPRING_2025_SKY16

M

MAT 110(Spring-26)11

M

MAT110_SUMMER_2025_SKY17

M

MAT110_FALL_2025_SKY04

M

MAT110_FALL_2025_SKY03

M

Mat-110-section-77

M

MAT110 Tutorial Spring 26All

M

MAT 121(Spring-26)01

M

MAT 110Section-18

E

EEE101SPRING20262

E

EEE101 || Spring-2026Section-7

M

MAT216 (SPRING 2026)Sec_15 & 16

E

EEE 101 / ECE 101 - Spring 2026Section 4

M

MAT110 (Spring 2026)Section-08 and Section-16

M

MAT216 Spring-26 22

M

MAT 120Section-03

M

MAT 1102

M

MAT316 Spring-26

M

MAT 110(Fall-25)18

M

MAT 110(Fall-25)20

M

MAT 122(Fall-25)01

Archived classes

Settings

Stream

Classwork

People

Sakal Roy

Mar 12, 2025

SO, EVEN IF YOU CAN'T JOIN FOR SOME REASON, MAKE SURE TO EMAIL TOPICS/SUBTOPICS/SPECIFIC PROBLEM NUMBER TO RECAP. AFTER FOCUSING ON THE OVERALL EXAM STRATEGY AND TOPICS, WE'LL FOCUS ON RECAPING BASED ON YOUR SUGGESTION. HOPE THAT HELPS. BEST WISHES.)

Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Sakal Roy is inviting you to a scheduled BdREN Zoom meeting.

Topic: MAT 110_SKY_EXAM REVIEW SESSION_1
Time: Mar 13, 2025 07:00 PM Astana, Dhaka

--> Join BdREN Zoom Meeting from Laptop or Mobile:
<https://bdren.zoom.us/j/2313081916?pwd=qdiZaZxXahPZ9k25qdixs9lbolJ0l9.1&omn=99210498015>

Meeting ID: 231 308 1916
Password: savelife

Add comment

Sakal Roy

Mar 12, 2025

Resources to look at for Differentiability and Techniques of differentiation.
There's more in the google doc. But these is the least that you have to go through to solve the practice problems.
Best wishes.

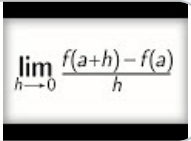
3.1 Derivative as slope

YouTube video • 7 minutes



3.2 Computing a derivative ...

YouTube video • 2 minutes



Introduction to Derivatives

<https://www.mathsisfun.com/calcul>



Derivative Rules

<https://www.mathsisfun.com/calcul>



Add comment

Sakal Roy

Mar 10, 2025

Hello, everyone.
At the very last moment, before joining you all in the class, I had to leave for a medical emergenc (not mine, I'm ok).
Couldn't even got the chance t to inform you here at last moment.
Extremely sorry for the inconvenience.

Now, the emergency is taken care of.
And, I will be available in my scheduled consultation hour (2:00 pm-3:30 pm) for today(MONDAY).
Drop by to clear your queries regarding limit, continuitiy.

Also, I'll announce the extra online class **date/time today based on your preferences in comment, which will make up also for todays lecture.**
See you all in next class.

Add comment

Sakal Roy

Mar 8, 2025

Announcement:

This week, I want to take an extra online problem solving session on limit, continuity and differentiability to wrap up thes topics once and for all. The session will be held on zoom and recording will be provided to you all.

So, please comment in this post regarding your favourable date and time for that extra online session.
I'll try to decide based on majoritys opinion.

These topics are the only place where you need to use limit and use it carefully.
For the rest of the topics for mid limit is not needed. Those are easier than limit solving cause they're most mostly focused on using directly the techniques of differentiation.

Remember, we've to make sure that everyone, I repeat everyone is clear about the practise problems in these topics. So, cooperate accordingly.
Best wishes.

Add comment

Sakal Roy

Mar 7, 2025

Let's welcome SADIA AFRIN
Student Tutor (ST):
Sec:16, 19, 20
Name: Sadia Afrin, Email: sadia.afrin9@g.bracu.ac.bd
I will request SADIA to share her consulation info when MNS finalizes it.
Hoping for a wonderful semester with everyones cooperation.

Add comment

Sakal Roy

Mar 5, 2025

https://classroom.google.com/u/8/c/NzU0Njg5ODY0MjY5

14/17

Team Quiz update.
Ques 4 and 5 will be given here in short.
See you all in the next class.

1) 10	10
2) 10	10
3) 10	10
4) To be	continued

Team Quiz Result_SEC_20.png

 Add comment



Sakal Roy
Mar 5, 2025

Announcement:
Today's class will officially start at **12:10 PM** with a short team quiz from the contents that we've already learned together. The first few minutes are for you all to form 2 teams by yourself and discuss/recap strategies between your teams. Then I'll start the class officially describing the team quiz rules.

Team Quiz Rules:

- 1) There'll be total 5 questions for each team. Questions may include MCQ questions, explanation, error finding, mathematical debate etc.
- 2) Team members can discuss the answer for 2 minutes within themselves after the question is revealed. But no resources except your reasoning are allowed during Team Quiz time.
- 3) Then in each round I'll choose 1 person from each team randomly to come to the whiteboard and answer/explain, debate the answer/explanation of that round's question.

Get ready. See the resources now. Let your sectionmates know about this in case they are not in this Google Classroom.

I'll start the Quiz at sharp 12:10 PM.

Best wishes.

Hints: see the Mathopolis questions on the posted links of the "Introduction to Limits" math is fun website resources.

 Add comment



Sakal Roy
Mar 4, 2025

SEE ATTACHED RESOURCE TO GET READY FOR THE NEXT CLASS.

MORE RESOURCES ARE IN THE GIVEN COURSE INFO GOOGLE DOC for SPRING 2025.

SEE YOU ALL. BEST WISHES.

<u>Limits to Infinity</u> https://www.mathsisfun.com/calcul	<u>Link</u> https://www.mathsisfun.com/calcul
<u>2.12 The Squeeze Theorem</u> YouTube video • 5 minutes	<u>6.15 Asymptotes</u> YouTube video • 5 minutes
<u>Continuous Functions</u> https://www.mathsisfun.com/calcul	<u>3.1 Derivative as slope</u> YouTube video • 7 minutes
<u>L'Hôpital's Rule</u> https://www.mathsisfun.com/calcul	<u>Course Info: MAT_110_SKY...</u> Google Docs

 Add comment



Sakal Roy

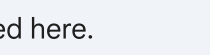
Mar 3, 2025

Introductory Limit resources are attached here.

These and more detailed resources are given for each topics in the course info google doc.

Limits (An Introduction)

<https://www.mathsisfun.com/calcul>



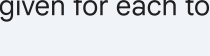
Calculus - The limit of a function...

YouTube video • 4 minutes



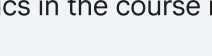
Limits Calculator Techniqu...

YouTube video • 14 minutes



What's Infinity?

<https://www.mathsisfun.com/numb>



 Add comment

 **Sakal Roy**
Mar 2, 2025

MAT110_SPRING2025_SKY

To share this document to anyone- USE the following link:

<https://rb.gy/Oa4xow>
or <https://docs.google.com/document/d/1LsnSjBDHpO7c7iYmUKxT2cep7AGiadSW9XqDvgcil20/>
or

SCAN the following QR Code:

You can share this link with the following foreword to anyone or Keep it confidential before the end of this semester.FOREWORD

Stream Classwork People



- Home
- Calendar
- Gemini
- Enrolled

^
- To-do
- M

MAT110_FALL2024_SKY

6
- M

MAT110_SPRING_2025_SKY

20
- M

MAT110_SPRING_2025_SKY

19
- M

MAT110_SPRING_2025_SKY

16
- M

MAT 110(Spring-26)

11
- M

MAT110_SUMMER_2025_SKY

17
- M

MAT110_FALL_2025_SKY

04
- M

MAT110_FALL_2025_SKY

03
- M

Mat-110-section-7

7
- M

MAT110 Tutorial Spring 26

All
- M

MAT 121(Spring-26)

01
- M

MAT 110

Section-18
- E

EEE101SPRING2026

2
- E

EEE101 || Spring-2026

Section-7
- M

MAT216 (SPRING 2026)

Sec_15 & 16
- E

EEE 101 / ECE 101 - Spring 2026

Section 4
- M

MAT110 (Spring 2026)

Section-08 and Section-16
- M

MAT216 Spring-26 22
- M

MAT 120

Section-03
- M

MAT 110

2
- M

MAT316 Spring-26
- M

MAT 110(Fall-25)

18
- M

MAT 110(Fall-25)

20
- M

MAT 122(Fall-25)

01
- Archived classes
- Settings

Comments are not a universal and complete guide for all learning objectives or learning documents. The reference books are better to refer to anyone for a complete view."

-

Sakal Roy
Adjunct (Contractual) Faculty,
MNS, BRACU,
Academic Counselor,
BdMOC,
Former Senior Math Olympiad Consultant,
PEDP4, DPE, MoPME,
Author, Secondary Mathematics Textbook,
NCTB
Member, Sub-committee:
Chess in School, Online and Digital Chess, Grassroot Chess Development,
Bangladesh Chess Federation



Add comment

Sakal Roy
Feb 28, 2025

FYI: MSG FROM Office of the Registrar, BRAC University=====

===Dear Students and Colleagues,

Ramadan Kareem to all of you!

Please see the class schedule and office timing during the holy month of Ramadan.

Class Schedule During the month of Ramadan – 2025

a. Undergraduate Programs:

<u>Current Timing_</u>	<u>Ramadan Timing</u>
08:00 AM - 09:20 AM	08:00 AM - 09:05 AM
09:30 AM - 10:50 AM	09:15 AM - 10:20 AM
11:00 AM - 12:20 PM	10:30 AM - 11:35 AM
12:30 PM - 01:50 PM	11:45 AM - 12:50 PM
02:00 PM - 03:20 PM	01:00 PM - 2:05 PM
03:30 PM - 04:50 PM	02:15 PM - 03:20 PM
05:00 PM - 06:20 PM	03:30 PM - 04:35 PM

b. Postgraduate Programs:

<u>Current Timing</u>	<u>Ramadan Timing</u>
06:30 PM - 09:30 PM	04:00 PM - 06:00 PM

Office Time During the month of Ramadan – 2025
09:00 AM - 3:30 PM

Kind regards,

Office of the Registrar
BRAC
University=====

===

Add comment

Sakal Roy
Feb 26, 2025

Dear ALL,
BRACU sugested to take totdays class online.
I'm assinging todays class as SELF-STUDY since after last class we all just need to go through the resources from "BEFORE CALCULUS" part on our own.
That means no further online class to attend today.

Add comment

Sakal Roy
Feb 25, 2025 (Edited Feb 25, 2025)

WELCOME TO MAT110_SPRING 2025_SKY_GROUP COMMUNICATION MEDIUM.
ASK YOUR SECTION MATES TO JOIN ASAP.
RESOURCES SHARED TILL NOW IN EMAIL ARE GIVEN HERE FOR CONVENIENCE.
EXPLORE AND ENJOY.

- MSG from Course Coordinator_MAT110_SPRING_2025:

=====

Hope this email finds you well. For your convenience, I have attached the course outline.

The mark distribution for MAT 110:
Attendance = 5%
Assignment = 15%
Quiz = 25%
Midterm = 20%

Final = 35%
Total = 100%

Stream Classwork People



Drive link for practice problems:

https://drive.google.com/drive/folders/1dHL4TyqNYR22brwj9XyDzFvGllkXbgvI?usp=drive_link

=====

- I'm also sharing my resources from the previous semester MAT 110 just to look at the full picture we need to cover unless BRACU, MNS makes any significant changes this semester.

MAT110_FALL2024_SKY

To share this document to anyone- USE the following link:

<https://shorturl.at/w6zWV>

I'll also provide another resource similar to this one containing this semester's final info provided by COURSE COORDINATOR, MAT110, SPRING 2025, MNS, BRACU. Till then we'll follow the contents from the above one.

=====

- **Fill up this form:**<https://forms.gle/EdXFY6u5LJzTpErq6>
-
- ***This is a diagnosis about what part of our mid term material we already know. But this one will not be graded, so I'm using an old one from my archive. Take time and fill up this form.***

=====

% My Consultation Info:

Consultation Room No:4 T 2 6 O

4 th Floor, MNS, Cubicle/Open Space for faculties

(1 st Row, Check My NAME and INITIAL: SKY)

Consultation Hour:

🕒 SAT : 2:00 PM - 3:30 PM

🕒 MON : 2:00 PM - 3:30 PM

Contact in my official /personal email IF: YOU DON;T SEE ME.

=====

Best regards

Sakal Roy

Academic Counselor & Associate Member, BdMOC

Former Senior Math Olympiad Consultant, MoPME

Author, Secondary Mathematics Textbook, NCTB

Adjunct (Contractual) Faculty,

Department of MNS, BRAC University

Member, Sub-committee: Chess in School, Online and Digital Chess, Grassroot Chess Development, Bangladesh Chess Federation

MAT110 Course Outline Spr...

Microsoft Word



Course Info: MAT_110_SKY_...

Google Docs



Google Forms: Sign-in

<https://forms.gle/EdXFY6u5LJzTpEr>



Add comment

- Home
- Calendar
- Gemini
- Enrolled ^
- To-do
- MAT110_FALL2024_SKY 6
- MAT110_SPRING_2025_SKY 20
- MAT110_SPRING_2025_SKY 19
- MAT110_SPRING_2025_SKY 16
- MAT 110(Spring-26) 11
- MAT110_SUMMER_2025_SKY 17
- MAT110_FALL_2025_SKY 04
- MAT110_FALL_2025_SKY 03
- Mat-110-section-7 7
- MAT110 Tutorial Spring 26 All
- MAT 121(Spring-26) 01
- MAT 110 Section-18
- EEE101SPRING2026 2
- EEE101 || Spring-2026 Section-7
- MAT216 (SPRING 2026) Sec_15 & 16
- EEE 101 / ECE 101 - Spring 2026 Section 4
- MAT110 (Spring 2026) Section-08 and Section-16
- MAT216 Spring-26 22
- MAT 120 Section-03
- MAT 110 2
- MAT316 Spring-26
- MAT 110(Fall-25) 18
- MAT 110(Fall-25) 20
- MAT 122(Fall-25) 01
- Archived classes
- Settings